Supplementary materials

Systematic Searches conducted December 2019 without publication date limits

PubMed

(Dementia [MH] OR dementia OR "cognitive dysfunction" [MH] OR "cognitive dysfunction" OR "cognitive impairment" OR "cognitive impairment" OR "cognitive decline" OR Alzheimer Disease [MH] OR Alzheimer* OR Vascular Dementia [MH] OR "Vascular dementia" OR "posterior cortical atrophy" OR Lewy Body Disease [MH] OR "lewy body disease" OR "lewy body dementia" OR "lewy bod*" OR "fronto-temporal dementia" OR "dementia with lewy bodies" OR "lewy body" OR "mixed dementia" OR Creutzfeldt-Jakob Syndrome [MH] OR "Creutzfeldt-Jakob" OR "Huntington's disease" OR "corticobasal degeneration" OR "senile dementia")

AND

Exercise [MH] OR exercise OR aerobic* OR running [MH] OR running OR walking [MH] OR walk* OR swimming [MH] OR swim* OR sports [MH] OR sport* OR bicycling [MH] OR bicycl* OR cycling OR "physical activity" OR Fitness OR Resistance training [MH] OR "resistance training" OR yoga [MH] OR yoga

- **Title and abstracts**
- **Limit to meta-analysis: 146**

Embase

Dementia OR cognitive dysfunction OR cognitive impairment OR cognitive decline OR Alzheimer Disease OR Alzheimer* OR Vascular dementia OR posterior cortical atrophy OR lewy body disease OR lewy body dementia OR lewy bod* OR fronto-temporal dementia OR dementia with lewy bodies OR lewy body OR mixed dementia OR Creutzfeldt-Jakob Syndrome OR Creutzfeldt-Jakob OR Huntingtons disease OR corticobasal degeneration OR senile dementia

AND

Exercise OR aerobic* OR running OR walking OR walk* OR swimming OR swim* OR sports OR sport* OR bicycling OR bicycl* OR cycling OR physical activity OR Fitness OR resistance training OR yoga

- **Title and abstracts**
- **Limit to meta-analysis and Medline: 51 hits**

PsychInfo

Dementia OR cognitive dysfunction OR cognitive impairment OR cognitive decline OR Alzheimer Disease OR Alzheimer* OR Vascular dementia OR posterior cortical atrophy OR lewy body disease OR lewy body dementia OR lewy bod* OR fronto-temporal dementia OR dementia with lewy bodies OR lewy body OR mixed dementia OR Creutzfeldt-Jakob Syndrome OR Creutzfeldt-Jakob OR Huntingtons disease OR corticobasal degeneration OR senile dementia

AND

Exercise OR aerobic* OR running OR walking OR walk* OR swimming OR swim* OR sports OR sport* OR bicycling OR bicycl* OR cycling OR physical activity OR Fitness OR resistance training OR yoga

Limit to meta-analysis: 77

Cochrane

(Dementia OR cognitive dysfunction OR cognitive impairment OR cognitive decline OR Alzheimer Disease OR Alzheimer* OR Vascular dementia OR posterior cortical atrophy OR lewy body disease OR lewy body dementia OR lewy bod* OR fronto-temporal dementia OR dementia with lewy bodies OR lewy body OR mixed dementia OR Creutzfeldt-Jakob Syndrome OR Creutzfeldt-Jakob OR Huntingtons disease OR corticobasal degeneration OR senile dementia) AND (Exercise OR aerobic* OR running OR walking OR walk* OR swimming OR swim* OR sports OR sport* OR bicycling OR bicycl* OR cycling OR physical activity OR Fitness OR resistance training OR yoga)

- **Title and abstracts**
- **Limit to reviews: 91 hits**

TOTAL: 365 citations

Table 4. Quality of meta-analyses according to AMSTAR-2 criteria

AMSTAR-2 criteria	**Angevaren *et al.*. 2008**	**Gates *et al.*. 2013**	**Heyn *et al.* 2004**	**Kelly *et al.*. 2014**	**Northey *et al.*. 2018**	**Sanders *et al* 2019**	**Scherder *et al.*. 2014**	**Song *et al.* 2010**	**Wang *et al.* 2015**	**Young *et al.* 2015**	**Zheng *et al* 2016**
Did the research question and inclusion criteria include components of PICO?	Y	Y	Y	Y	Y	Y	N	Y	N	Y	Y
Were the review methods were established prior to the conduct of the review?	N	N	N	Y	N	Y	N	N	N	N	Y
Did the authors explain their selection of the study designs for inclusion?	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Was a comprehensive literature search strategy used?	Y	Y	Y	Y	Y	Y*	Y*	Y*	Y	Y	Y
Was study selection performed in duplicate?	Y	N	N	Y	Y	Y	Y	Y	N	Y	Y
Was data extraction performed in duplicate?	Y	Y	N	Y	Y	N	N	Y	Y	Y	Y
Was a list of excluded studies and justification for exclusions provided?	Y	N	N	Y	Y	Y*	N	Y	Y	Y	Y*
Were the included studies described in detail?	Y	Y	Y*	Y	Y	Y*	Y	Y	Y	Y	Y*
Was a satisfactory technique used to assess RoB?	Y	Y	Y	Y	Y	Y	N	Y	Y	Y	Y
Were sources of funding of the included studies reported?	N	N	N	N	N	Y	N	N	N	N	Y
Were appropriate methods used to statistical combine results for the meta-analyses?	Y	Y	N	Y	Y	N	Y	Y	Y	Y	Y
Was potential impact of RoB in individual studies on the results of the meta-analyses assessed?	N	N	Y	N	Y	Y	N	Y	Y	Y	N
Was RoB in individual studies accounted for when interpreting/discussing the results?	N	N	Y	N	Y	Y	N	Y	Y	Y	Y
Was a satisfactory explanation for, and discussion of, any heterogeneity observed in the results provided?	N	Y	N	Y	N	N	N	Y	Y	N	Y
Was an adequate investigation of publication bias carried out?	N	N	N	N	Y	Y	Y	N	N	N	Y
Did the authors report any potential sources of conflict of interest?	N	Y	Y	Y	Y	Y	Y	N	Y	Y	Y
AMSTAR 2 rating	**Critically low**	**Moderate**	**Low**	**Low**	**Moderate**	**Low**	**Critically low**	**Moderate**	**Moderate**	**Moderate**	**High**

Note. * denotes partial yes or no.